

Logo
Not
Available



Qualification Pack



SENIOR AUTOMATION TECHNICIAN

QP Code: CRI/ELE/Q0801

Version: 1.0

NSQF Level: 4.5

CRISP Society || CRISP, Opposite Manas Bhawan, Shyamla Hills
Bhopal || email: sudheer@crispindia.com

Qualification Pack

Contents

CRI/ELE/Q0801: SENIOR AUTOMATION TECHNICIAN	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CRI/ELE/N0801: Applications of PLC Programming- Architecture and Design	5
CRI/ELE/N0802: Control Technologies in Automation- Man, Machine Interface (SCADA)	14
CRI/ELE/N0803: System Design and Integration	21
DGT/VSQ/N0102: Employability Skills (60 Hours)	28
Assessment Guidelines and Weightage	35
<i>Assessment Guidelines</i>	35
<i>Assessment Weightage</i>	36
Acronyms	37
Glossary	38

Qualification Pack

CRI/ELE/Q0801: SENIOR AUTOMATION TECHNICIAN

Brief Job Description

A Senior Automation Technician is a professional responsible for designing, implementing, and maintaining automation systems in industrial settings. Their primary goal is to optimize processes, enhance efficiency, and ensure the smooth operation of manufacturing or production facilities. The job description may vary depending on the industry and specific company needs but generally includes the following responsibilities and Technical & Managerial skills.

Personal Attributes

Honesty, tech savvy, technically sound, self motivated, upgraded with updated technology, adaptive, efficient and should have sound knowledge about the recent updations in the automation segment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CRI/ELE/N0801: Applications of PLC Programming- Architecture and Design](#)
2. [CRI/ELE/N0802: Control Technologies in Automation- Man, Machine Interface \(SCADA\)](#)
3. [CRI/ELE/N0803: System Design and Integration](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Strategic Electronics
Occupation	Programmer
Country	India
NSQF Level	4.5
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 7412.0101

Qualification Pack

Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th (Relevant field) with NA of experience OR Pursuing 3rd year of 3-year diploma after 10th (Relevant field) with NA of experience OR 10th grade pass with 2 year NTC plus 1 year NAC (Relevant field) with 2 Years of experience OR 12th grade Pass (Graduation in Engineering) with 3.5 years of experience Graduation in Engineering
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-4.5-EH-02556-2024-V1-CRISP
NQR Version	1.0

Qualification Pack

CRI/ELE/N0801: Applications of PLC Programming- Architecture and Design

Description

The primary guiding force driving automated industrial plants is a "Programmable Logic Controller," commonly referred to as a PLC. A PLC serves as a computing system strategically programmed to oversee electromechanical processes within industries. This unit provides an understanding of both conceptual and practical fundamentals of automation, encompassing PLC hardware and programming. The critical attributes of PLC hardware and programming play pivotal roles in the realm of industrial automation.

Scope

The scope covers the following :

- Understand the concept of Automation in Industry
- Understand the basics of PLC.
- Familiar with Hardware of PLC
- Use of Totally Integrated Automation Portal (TIA) for PLC Programming.
- Check out the basic hardware connection of PLC.

Elements and Performance Criteria

Understand the concept of Automation in Industry

To be competent, the user/individual on the job must be able to:

- PC1.** Explain use of automation for industrial safety
- PC2.** Describe levels of Industrial Automation Process
- PC3.** State the advantages & disadvantages of Industrial Automation

Understanding the concept of Programmable Logic Control (PLC)

To be competent, the user/individual on the job must be able to:

- PC4.** Describe the Industrial Application of PLC
- PC5.** Explain the Basic Architecture of PLC System
- PC6.** Identify the different types of PLC available in lab
- PC7.** Explain PLC Scan Cycle and Scan time
- PC8.** Explain various communication protocol of PLC

Understanding Programmable Logic Control (PLC) selection criteria

To be competent, the user/individual on the job must be able to:

- PC9.** Describe different Programmable Logic Control (PLC) selection criteria
- PC10.** Identify which PLC is suitable for given application

Understanding PLC Hardware (Siemens Family)

To be competent, the user/individual on the job must be able to:

- PC11.** Identify the various PLC hardware like SIMATIC S7 family
- PC12.** Numbers of modules can connect with S71200 PLC
- PC13.** Explain the use of Signal Modules

Qualification Pack

Understanding Concept of Memory Organization and I/O addressing

To be competent, the user/individual on the job must be able to:

- PC14.** What are the different data types use for PLC?
- PC15.** Explain the use of PLC tags
- PC16.** Identify how many PLC tags are required for given Example

Understanding concept of NO/NC (Normally Open/ Normally Closed) for PLC

To be competent, the user/individual on the job must be able to:

- PC17.** Explain NO/NC concept
- PC18.** Explain relay concept
- PC19.** Identify the use of NO/NC

Describe concept of Ladder logic for PLC Programming

To be competent, the user/individual on the job must be able to:

- PC20.** Explain important rules of ladder logic
- PC21.** Identify different Logic Gates
- PC22.** Identify different types of gates
- PC23.** Draw ladder logic for all gates
- PC24.** Explain concept of latching
- PC25.** Explain concept of Interlocking

Understanding Totally Integrated Automation Portal (TIA) for PLC Programming

To be competent, the user/individual on the job must be able to:

- PC26.** Explain the different TIA license
- PC27.** Create project in TIA portal
- PC28.** Creating new PLC project in TIA
- PC29.** Assign IP address to the PLC
- PC30.** Add available Hardware Modules in TIA
- PC31.** Downloading new project in Hardware of PLC using Ethernet cable
- PC32.** Identify the various Indication on Hardware

Use of Main OB in Totally Integrated Automation Portal (TIA) for PLC Programming

To be competent, the user/individual on the job must be able to:

- PC33.** Draw the simple NO/NC ladder program in main OB and execute
- PC34.** Draw ladder diagram for all logic gate and execute
- PC35.** Draw the ladder diagram for latching example using push button and execute
- PC36.** Draw the ladder for interlocking example and execute
- PC37.**
 - Draw the ladder diagram for home
 - automation for Example “automatic door” or
 - “automatic fan “& execute
- PC38.**
 - Create own tag table using various data
 - types of PLC

Use different Comparison and arithmetic instruction in Totally Integrated Automation Portal (TIA) for PLC Programming

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC39.** Explain different comparison instruction
- PC40.**
- Draw ladder diagram using comparison
 - instruction (example)
- PC41.**
- Draw ladder diagram using arithmetic
 - instruction (example)
- PC42.**
- Design calculator using arithmetic
 - instruction and execute

Use different Timer and Counter instruction in Totally Integrated Automation Portal (TIA) for PLC Programming

To be competent, the user/individual on the job must be able to:

- PC43.** Explain different timer instruction
- PC44.**
- Draw ladder diagram using timer instruction
 - (example)
- PC45.** Explain different counter instruction
- PC46.**
- Draw ladder diagram using counter
 - instruction (example)
- PC47.**
- Draw ladder diagram for “Traffic signal “and
 - execute
- PC48.**
- Draw ladder diagram for “Smart Car Parking
 - “and execute

Creating Standard Function in Totally Integrated Automation Portal (TIA) for PLC Programming

To be competent, the user/individual on the job must be able to:

- PC49.** Explain use of function in PLC
- PC50.** Create one function in TIA and execute
- PC51.**
- Draw ladder diagram for “Machine
 - Simulation“ using function and execute

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Hardware use in Industrial Automation
- KU2.** PLC types and applications
- KU3.** Selection criteria of PLC
- KU4.** Communication protocol
- KU5.** Latest technologies used in system configurations
- KU6.** Communicate PLC with PC using TIA
- KU7.** Various license of TIA portal
- KU8.** Indication on PLC like error, run, maintenance
- KU9.** Connection of PLC with modules
- KU10.** Rules of ladder logic programming
- KU11.** Basic ladder logic programming

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Read vernacular/English language
- GS2.** Read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** Communicate with Supervisor
- GS4.** Follow organization rule-based decision-making process
- GS5.** Take decision with systematic course of actions and/or response
- GS6.** Organize work to meet deadlines
- GS7.** Work constructively and collaboratively with others
- GS8.** Follow code of conduct
- GS9.** Manage relationships with customers with intent on satisfying requirements of service delivery
- GS10.** Recognize problems and search for solutions GS16. choose best methods to complete assigned tasks

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand the concept of Automation in Industry</i>	3	2	-	-
PC1. Explain use of automation for industrial safety	-	-	-	-
PC2. Describe levels of Industrial Automation Process	-	-	-	-
PC3. State the advantages & disadvantages of Industrial Automation	-	-	-	-
<i>Understanding the concept of Programmable Logic Control (PLC)</i>	4	4	-	2
PC4. Describe the Industrial Application of PLC	-	-	-	-
PC5. Explain the Basic Architecture of PLC System	-	-	-	-
PC6. Identify the different types of PLC available in lab	-	-	-	-
PC7. Explain PLC Scan Cycle and Scan time	-	-	-	-
PC8. Explain various communication protocol of PLC	-	-	-	-
<i>Understanding Programmable Logic Control (PLC) selection criteria</i>	2	2	-	2
PC9. Describe different Programmable Logic Control (PLC) selection criteria	-	-	-	-
PC10. Identify which PLC is suitable for given application	-	-	-	-
<i>Understanding PLC Hardware (Siemens Family)</i>	2	4	-	2
PC11. Identify the various PLC hardware like SIMATIC S7 family	-	-	-	-
PC12. Numbers of modules can connect with S71200 PLC	-	-	-	-
PC13. Explain the use of Signal Modules	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Concept of Memory Organization and I/O addressing</i>	3	6	-	-
PC14. What are the different data types use for PLC?	-	-	-	-
PC15. Explain the use of PLC tags	-	-	-	-
PC16. Identify how many PLC tags are required for given Example	-	-	-	-
<i>Understanding concept of NO/NC (Normally Open/ Normally Closed) for PLC</i>	2	4	-	-
PC17. Explain NO/NC concept	-	-	-	-
PC18. Explain relay concept	-	-	-	-
PC19. Identify the use of NO/NC	-	-	-	-
<i>Describe concept of Ladder logic for PLC Programming</i>	3	8	-	2
PC20. Explain important rules of ladder logic	-	-	-	-
PC21. Identify different Logic Gates	-	-	-	-
PC22. Identify different types of gates	-	-	-	-
PC23. Draw ladder logic for all gates	-	-	-	-
PC24. Explain concept of latching	-	-	-	-
PC25. Explain concept of Interlocking	-	-	-	-
<i>Understanding Totally Integrated Automation Portal (TIA) for PLC Programming</i>	3	10	8	2
PC26. Explain the different TIA license	-	-	-	-
PC27. Create project in TIA portal	-	-	-	-
PC28. Creating new PLC project in TIA	-	-	-	-
PC29. Assign IP address to the PLC	-	-	-	-
PC30. Add available Hardware Modules in TIA	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. Downloading new project in Hardware of PLC using Ethernet cable	-	-	-	-
PC32. Identify the various Indication on Hardware	-	-	-	-
<i>Use of Main OB in Totally Integrated Automation Portal (TIA) for PLC Programming</i>	2	8	6	2
PC33. Draw the simple NO/NC ladder program in main OB and execute	-	-	-	-
PC34. Draw ladder diagram for all logic gate and execute	-	-	-	-
PC35. Draw the ladder diagram for latching example using push button and execute	-	-	-	-
PC36. Draw the ladder for interlocking example and execute	-	-	-	-
PC37. • Draw the ladder diagram for home • automation for Example “automatic door” or • “automatic fan “& execute	-	-	-	-
PC38. • Create own tag table using various data • types of PLC	-	-	-	-
<i>Use different Comparison and arithmetic instruction in Totally Integrated Automation Portal (TIA) for PLC Programming</i>	2	6	6	2
PC39. Explain different comparison instruction	-	-	-	-
PC40. • Draw ladder diagram using comparison • instruction (example)	-	-	-	-
PC41. • Draw ladder diagram using arithmetic • instruction (example)	-	-	-	-
PC42. • Design calculator using arithmetic • instruction and execute	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Use different Timer and Counter instruction in Totally Integrated Automation Portal (TIA) for PLC Programming</i>	2	8	4	3
PC43. Explain different timer instruction	-	-	-	-
PC44. • Draw ladder diagram using timer instruction • (example)	-	-	-	-
PC45. Explain different counter instruction	-	-	-	-
PC46. • Draw ladder diagram using counter • instruction (example)	-	-	-	-
PC47. • Draw ladder diagram for “Traffic signal “and • execute	-	-	-	-
PC48. • Draw ladder diagram for “Smart Car Parking • “and execute	-	-	-	-
<i>Creating Standard Function in Totally Integrated Automation Portal (TIA) for PLC Programming</i>	2	8	6	3
PC49. Explain use of function in PLC	-	-	-	-
PC50. Create one function in TIA and execute	-	-	-	-
PC51. • Draw ladder diagram for “Machine • Simulation“ using function and execute	-	-	-	-
NOS Total	30	70	30	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CRI/ELE/N0801
NOS Name	Applications of PLC Programming- Architecture and Design
Sector	Electronics
Sub-Sector	
Occupation	Programmer, Programmer
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

CRI/ELE/N0802: Control Technologies in Automation- Man, Machine Interface (SCADA)

Description

As industries continue to evolve towards increased automation, the integration of a Supervisory Control and Data Acquisition (SCADA) system becomes imperative. The SCADA system acts as the vital link between human operators and the PLC, facilitating real-time monitoring, control, and optimization of industrial processes. This rationale explores the key attributes and significance of the Man-Machine Interface (MMI) in the form of SCADA within automated industrial plants.

Scope

The scope covers the following :

- Receive an overview of the scope of performance of the engineering framework.
- Understand the concept of SCADA.
- Design multiple screens for the visualization of automation process.
- Use of Totally Integrated Automation Portal (TIA) for SCADA designing.
- Familiar with the system requirements.

Elements and Performance Criteria

Understanding the need of SCADA

To be competent, the user/individual on the job must be able to:

PC1. Explain use of automation for industrial safety

PC2. Explain level of industrial automation

PC3. What are the advantages & disadvantage of industrial Automation

Planning the structure of the process display for SCADA

To be competent, the user/individual on the job must be able to:

PC4. Describe SCADA design Plan for the given example

PC5. Identify operations/input are permitted to operator for given example

Understanding Totally Integrated Automation Portal (TIA) for SCADA Design

To be competent, the user/individual on the job must be able to:

PC6. Create new Project for SCADA Design in TIA

PC7. Configuring the communication modules of SCADA in TIA

PC8. Create communication between the PLC and SCADA system

PC9. Add new screen in project for SCADA in TIA

PC10. Simulating the runtime of SCADA

Understanding Tag Management for SCADA Design

To be competent, the user/individual on the job must be able to:

PC11. Describe difference between internal and external tags

PC12. Explain power tags

PC13. Create and linking tags with PLC

Qualification Pack

Designing of graphic screens

To be competent, the user/individual on the job must be able to:

PC14. Create or add new screen and change it as main screen in TIA SCADA

PC15. Configure different basic object on Screen of SCADA and execute

PC16. Configure different basic element on Screen of SCADA and execute

PC17. Configure different basic control on Screen of SCADA and execute

Understanding different Events and Functions

To be competent, the user/individual on the job must be able to:

PC18. Configure button on Screen with bit event and execute

PC19. Configure output object on Screen with bit function and execute

PC20. • Change the various properties of button such as Colour, shape, appearance of button on Screen and
• execute

PC21. • Change the various properties of output object such as Colour, shape, appearance of button on
• Screen and execute

Use of Animation for object and elements

To be competent, the user/individual on the job must be able to:

PC22. Apply the various animation on button such as appearance, display of button on Screen and execute

PC23. • Apply the various animation on output object such as appearance, display of button on Screen and
• execute

PC24. Apply various movement animation on Object and execute

PC25. Design “Conveyer Belt “movement operation on Screen and execute

Use of Controls like Trends, Alarm

To be competent, the user/individual on the job must be able to:

PC26. Design Trend control on Screen and execute

PC27. Design alarm control on Screen and execute

PC28. Design complete SCADA screen for any Industrial Application (Ex. Sorting Material)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Human machine interface important

KU2. SCADA and its applications

KU3. Design criteria of SCADA

KU4. Communication protocol

KU5. Latest technologies used in system configurations

KU6. Communicate PLC with SCADA system using TIA

KU7. Various license of TIA portal

KU8. Update in SCADA design

KU9. Connection of PLC with modules

Qualification Pack

KU10. Rules of SCADA design

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read vernacular/English language
- GS2.** Read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** Communicate with Supervisor
- GS4.** Follow organization rule-based decision-making process
- GS5.** Take decision with systematic course of actions and/or response
- GS6.** Organize work to meet deadlines
- GS7.** Work constructively and collaboratively with others
- GS8.** Follow code of conduct
- GS9.** Manage relationships with customers with intent on satisfying requirements of service delivery
- GS10.** Recognize problems and search for solutions GS16. choose best methods to complete assigned tasks

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the need of SCADA</i>	6	-	-	-
PC1. Explain use of automation for industrial safety	-	-	-	-
PC2. Explain level of industrial automation	-	-	-	-
PC3. What are the advantages & disadvantage of industrial Automation	-	-	-	-
<i>Planning the structure of the process display for SCADA</i>	4	2	-	-
PC4. Describe SCADA design Plan for the given example	-	-	-	-
PC5. Identify operations/input are permitted to operator for given example	-	-	-	-
<i>Understanding Totally Integrated Automation Portal (TIA) for SCADA Design</i>	2	8	4	2
PC6. Create new Project for SCADA Design in TIA	-	-	-	-
PC7. Configuring the communication modules of SCADA in TIA	-	-	-	-
PC8. Create communication between the PLC and SCADA system	-	-	-	-
PC9. Add new screen in project for SCADA in TIA	-	-	-	-
PC10. Simulating the runtime of SCADA	-	-	-	-
<i>Understanding Tag Management for SCADA Design</i>	4	4	3	4
PC11. Describe difference between internal and external tags	-	-	-	-
PC12. Explain power tags	-	-	-	-
PC13. Create and linking tags with PLC	-	-	-	-
<i>Designing of graphic screens</i>	2	16	-	4

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Create or add new screen and change it as main screen in TIA SCADA	-	-	-	-
PC15. Configure different basic object on Screen of SCADA and execute	-	-	-	-
PC16. Configure different basic element on Screen of SCADA and execute	-	-	-	-
PC17. Configure different basic control on Screen of SCADA and execute	-	-	-	-
<i>Understanding different Events and Functions</i>	3	14	-	-
PC18. Configure button on Screen with bit event and execute	-	-	-	-
PC19. Configure output object on Screen with bit function and execute	-	-	-	-
PC20. • Change the various properties of button such as Colour, shape, appearance of button on Screen and • execute	-	-	-	-
PC21. • Change the various properties of output object such as Colour, shape, appearance of button on • Screen and execute	-	-	-	-
<i>Use of Animation for object and elements</i>	3	14	7	4
PC22. Apply the various animation on button such as appearance, display of button on Screen and execute	-	-	-	-
PC23. • Apply the various animation on output object such as appearance, display of button on Screen and • execute	-	-	-	-
PC24. Apply various movement animation on Object and execute	-	-	-	-
PC25. Design “Conveyer Belt “movement operation on Screen and execute	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Use of Controls like Trends, Alarm</i>	6	12	16	6
PC26. Design Trend control on Screen and execute	-	-	-	-
PC27. Design alarm control on Screen and execute	-	-	-	-
PC28. Design complete SCADA screen for any Industrial Application (Ex. Sorting Material)	-	-	-	-
NOS Total	30	70	30	20

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CRI/ELE/N0802
NOS Name	Control Technologies in Automation- Man, Machine Interface (SCADA)
Sector	Electronics
Sub-Sector	
Occupation	Programmer, Programmer
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

CRI/ELE/N0803: System Design and Integration

Description

The system design ensures compliance with industry regulations and safety standards. PLC programming includes safety interlocks and emergency shutdown features to mitigate risks. and also includes diagnostics and error-handling features, enhancing the overall reliability and minimizing downtime. The well-thought-out design and integration strategies contribute to the efficiency and success of the automated industrial processes. This unit is about analyzing and understanding user requirements, designing the system and testing it to conform to the user/client specifications.

Scope

The scope covers the following :

- The scope covers the following :
- The system comprises various components of diverse types, encompassing electrical components, electronic assemblies, instrumentation and sensors, electro-mechanical devices, pneumatic and hydraulic
- components, wiring, control panels, PLCs, SCADA, communication components, and associated software.
- This unit/task encompasses the following:
- Analyzing and understanding client/user requirements
- Design and Develop control Systems
- Testing the system develop

Elements and Performance Criteria

Analyse client/user requirements

To be competent, the user/individual on the job must be able to:

- PC1.**
 - Identify the automation requirements in their
 - industrial process /control system
- PC2.**
 - Explain the significance of the possible
 - automation in the existing processes with emphasis
 - on latest practices in automation
- PC3.**
 - Assist the client/user in selection of suitable
 - automation systems/ sub systems as per their
 - existing processes
- PC4.**
 - Suggest technical specification of identified
 - systems / sub systems for automation based on
 - project requirements
- PC5.**
 - Formulate a concise summary of the user
 - requirement specifications and validate
 - comprehension with the client.

Design and Develop control Systems

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC6.**
- Design a control system application in accordance with user requirements, adhering to the organization's standard operating procedure (SOP)
- PC7.**
- Use accepted engineering concepts, processes, and principles to develop the control panel application
- PC8.**
- Identify the requirement of indications, switchgears and accessories etc
- PC9.**
- Prepare general arrangement diagram including wiring and loading plans
- PC10.**
- Integrate the diverse process system, sub systems and components as per the user requirement (e.g, using communication protocol)
- PC11.**
- Ensure seamless communication and interoperability among various devices and software
- PC12.**
- Identifying potential risks associated with the design and development of control systems
- PC13.**
- Compliance considerations into system design to meet legal and safety requirements
- PC14.**
- Plan and execute effective strategies to ensuring system reliability and safety

Testing the system develop

To be competent, the user/individual on the job must be able to:

- PC15.**
- Diagnose issues that may arise during the development
- PC16.**
- Diagnose issues that may arise during the development
- PC17.** Test the system in off line mode using simulator
- PC18.**
- Verify that the system conforms with all the industry regulations and standards relevant to industrial
- PC19.**
- Rework if there are any issues found and fix them
- PC20.**
- Test for integration of main system with the sub-systems (if applicable)
- PC21.** Ensure efficient and error-free operation
- PC22.** Perform Factory Acceptance Test (FAT)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's code of conduct
- KU2.** Organization culture
- KU3.** Company's reporting structure
- KU4.** Company's documentation policy

Qualification Pack

- KU5.** Company's line of business and product offerings
- KU6.** Company's production policy
- KU7.** Departments involved with production
- KU8.** Quality and standards system followed in the Electrical, electronics and instrumentation
- KU9.** Basics of computers and human machine interface

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read vernacular/English language
- GS2.**
 - read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** communicate with Supervisor
- GS4.** follow organization rule-based decision-making process
- GS5.** take decision with systematic course of actions and/or response
- GS6.** organize work to meet deadlines
- GS7.** work constructively and collaboratively with others
- GS8.** follow code of conduct
- GS9.**
 - manage relationships with customers with intent on satisfying requirements of service
 - delivery
- GS10.**
 - recognize problems and search for solutions
- GS16.** choose best methods to complete assigned tasks

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyse client/user requirements</i>	8	16	8	4
PC1. • Identify the automation requirements in their • industrial process /control system	-	-	-	-
PC2. • Explain the significance of the possible • automation in the existing processes with emphasis • on latest practices in automation	-	-	-	-
PC3. • Assist the client/user in selection of suitable • automation systems/ sub systems as per their • existing processes	-	-	-	-
PC4. • Suggest technical specification of identified • systems / sub systems for automation based on • project requirements	-	-	-	-
PC5. • Formulate a concise summary of the user • requirement specifications and validate • comprehension with the client.	-	-	-	-
<i>Design and Develop control Systems</i>	12	34	12	10
PC6. • Design a control system application in • accordance with user requirements, adhering to the • organization's standard operating procedure (SOP)	-	-	-	-
PC7. • Use accepted engineering concepts, processes, • and principles to develop the control panel • application	-	-	-	-
PC8. • Identify the requirement of indications, • switchgears and accessories etc	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. • Prepare general arrangement diagram including • wiring and loading plans	-	-	-	-
PC10. • Integrate the diverse process system, sub • systems and components as per the user • requirement (e.g, using communication protocol)	-	-	-	-
PC11. • Ensure seamless communication and • interoperability among various devices and software	-	-	-	-
PC12. • Identifying potential risks associated with the • design and development of control systems	-	-	-	-
PC13. • Compliance considerations into system design • to meet legal and safety requirements	-	-	-	-
PC14. • Plan and execute effective strategies to • ensuring system reliability and safety	-	-	-	-
<i>Testing the system develop</i>	10	20	10	6
PC15. • Diagnose issues that may arise during the • development	-	-	-	-
PC16. • Diagnose issues that may arise during the • development	-	-	-	-
PC17. Test the system in off line mode using simulator	-	-	-	-
PC18. • Verify that the system conforms with all the • industry regulations and standards relevant to • industrial	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. • Rework if there are any issues found and fix them	-	-	-	-
PC20. • Test for integration of main system with the sub-systems (if applicable)	-	-	-	-
PC21. Ensure efficient and error-free operation	-	-	-	-
PC22. Perform Factory Acceptance Test (FAT)	-	-	-	-
NOS Total	30	70	30	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CRI/ELE/N0803
NOS Name	System Design and Integration
Sector	Electronics
Sub-Sector	
Occupation	Programmer, Programmer
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Logo
Not
Available



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

Qualification Pack

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CRI/ELE/N0801.Applications of PLC Programming- Architecture and Design	30	70	30	20	150	35
CRI/ELE/N0802.Control Technologies in Automation- Man, Machine Interface (SCADA)	30	70	30	20	150	25
CRI/ELE/N0803.System Design and Integration	30	70	30	20	150	25
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	110	240	90	60	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
HMI	Human-Machine Interface
SCADA	Supervisory control and data acquisition
PLC	Programmable Logic Controller

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.